

Memory by Collapse: One-Fold Writing, Settling Recall, and the Direction of Thought

Engines 18–24 of the NEXUS Program

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Abstract

Contemporary artificial intelligence is built on an expansion primitive: the feedforward pass takes one input and unfolds it through billions of parameters, learning by global gradient descent over a frozen architecture. This paper argues, and then measures, the inverse thesis of the NEXUS framework: that thought is a collapse operation — many into one — and that a memory system built on collapse rather than expansion exhibits, at matched parameter budget, three properties the expansion architecture structurally cannot have: writing that never erases, learning whose cost is a single local fold, and recall by constraint-settling from partial cues. We instantiate the smallest complete such system (the Nexus Cell, a constraint-settling associative memory in classical and sharpened forms), race it against a parameter-matched feedforward network (the sled) on a continual stream protocol with no replay, and report all results from live runs, including the negatives. The cell holds its earliest memories at full strength after a thousand subsequent one-shot writes (100% recall, first-written = last-written) while the matched feedforward network exhibits total catastrophic forgetting (0% on early items). Writing costs the cell approximately 256 operations per memory against approximately 3.1×10^8 multiply-accumulates for the gradient-trained network — a ratio of roughly six orders of magnitude at equal parameter count in the classical comparison (5,760× measured directly). We further report two honest negative results: the classical cell's catastrophic capacity wall at $0.138N$ (which the sharpened cell removes), and the failure, under two separate dynamical protocols, of the hypothesis that the harmonic constant $H = \pi/g$ is a universal feedback optimum — the measured law in deep residual networks is $\alpha^* \approx 2.5/\sqrt{L}$. Finally, we note the published identity between the sharpened cell's settling step and transformer attention, which locates the collapse operation inside the very architecture the thesis criticizes, and we situate these results alongside the author's v55 Triadic Cell finding that compiled structure lifted a 1.5B-parameter model from 0.906 to 1.000 on an adversarial benchmark. The components used here are established in the literature; the contribution is the constraint-first synthesis, the matched-budget stream measurements, the energy ledger, and the preserved negatives.

1. Introduction: The Direction of the Arrow

The NEXUS framework holds that the universe runs an expansion under pressure — one becomes many — while thought runs the inverse: many becomes one, collapsing back toward a few or a single resolved structure. Read as engineering rather than metaphor, this is a claim about which primitive an artificial intelligence should be built on. The dominant architecture, the transformer, is an expansion engine: each forward pass widens the representation, capacity is bought with parameters, learning requires a global backward pass over a frozen network, and the context stack fills until it maxes out. The framework predicts that an architecture whose native operation is collapse — settling a state under constraints until it closes on the structure the input already implied — should learn locally, remember without erasing, and operate at a small fraction of the energy.

Two prior observations anchor the claim. First, BBP-type formulas read a hexadecimal digit of π at an arbitrary position without computing the preceding digits: the answer is addressed, not constructed. Second, SHA-256 derives a structural fingerprint of an input from the input itself in a fixed number of folds. Both are cheap because they do not search; they let structure that is already present fall into view. The hypothesis of this paper is that memory and recall can be built the same way: writing as a single fold that deposits constraint, recall as a settling — clamp what is known and let the unknown adjust until the configuration closes, the way one fixes C and adjusts a and b to satisfy it.

The paper follows the NEXUS engine discipline throughout: every number reported below is from a live, seeded run; protocols that produced results contrary to the framework's expectations are reported with the same prominence as the confirmations; and components inherited from prior literature are credited as such, with the contribution of this work stated narrowly.

2. Provenance and Related Work

Nothing in the machinery below is invented here, and the paper's credibility rests on saying so plainly. The classical constraint-settling memory is the Hopfield network (Hopfield 1982): symmetric couplings, Hebbian one-shot storage, recall by asynchronous descent to an attractor. Its capacity limit of approximately $0.138N$ patterns, and the catastrophic (rather than graceful) failure beyond it, are classical results (Amit, Gutfreund and Sompolinsky 1985). Catastrophic forgetting in gradient-trained feedforward networks is likewise long established (McCloskey and Cohen 1989; French 1999), as is a literature of mitigations (rehearsal, elastic weight consolidation) that this paper does not benchmark against and does not claim to supersede. The sharpened cell used in Engine 24 is the dense associative memory of Krotov and Hopfield (2016), whose modern continuous form was shown by Ramsauer et al. (2020) to have an update rule mathematically identical to transformer attention — an identity this paper uses interpretively but did not discover. The thermodynamic framing — that erasure, not computation, is what costs energy — is Landauer (1961). The variational reading of physics invoked in the discussion (laws as constraint closure, conservation as the shadow of symmetry) is the standard Lagrangian/Noether formulation.

Against that inherited base, the specific contributions of this paper are: (i) the constraint-first synthesis — assembling these components as the minimal instantiation of the NEXUS collapse thesis, with each design choice forced by a stated constraint rather than searched; (ii) matched-parameter, head-to-head stream measurements of one-fold writing against sequential gradient training with no replay, including first-written versus last-written retention; (iii) an explicit energy ledger in multiply-accumulate operations for learning under both primitives; (iv) an empirical scaling boundary for fragment-quality versus library size in the

sharpened cell; and (v) two preserved negative results concerning the $H = \pi/9$ universal-attractor hypothesis under dynamical protocols, which constrain where in the theory that constant can live.

3. Methods

3.1 The Nexus Cell

Classical form (Engine 23). $N = 256$ bipolar units; symmetric coupling matrix W with zero diagonal; 32,640 free couplings. Writing a memory x is one local Hebbian fold, $W \leftarrow W + (1/N) \cdot xx^T$, costing $N^2 = 65,536$ multiply-accumulates and requiring no gradient, no freeze, and no access to previously stored patterns. Recall clamps a fragment as the initial state and relaxes by asynchronous sign updates (up to ten random-order sweeps) until no unit flips. A fragment with keep-fraction f retains $f \cdot N$ true bits; the remainder are random ± 1 noise, the harder and fairer cue condition. Recall is scored by absolute overlap $m = |s \cdot x|/N$, with success defined as $m > 0.9$.

Sharpened form (Engine 24). The same units, but memories are stored as rows of a pattern matrix X and recall is the settling iteration $s \leftarrow X^T \text{softmax}(\beta X s)$ with $\beta = 0.2$, iterated to convergence (empirically two to three steps). Writing is an append: one row, approximately N operations, with zero disturbance of any prior memory. This is the dense associative memory of Krotov–Hopfield in the continuous form of Ramsauer et al.; its single settling step is the attention operation. Storage grows linearly with the number of memories — under the framework this is the point (memory grows from the query) rather than a defect, but the recall cost $K \cdot N$ per step grows with library size and is reported as such.

3.2 The Sled (matched baseline)

A feedforward autoencoder, 256–64–256 with tanh activations: 32,768 weights, matching the classical cell's 32,640 couplings to within 0.4%. Training is stochastic gradient descent with momentum (learning rate 0.05, momentum 0.9) on the same fragment-to-pattern reconstruction task the cell is tested on, so the baseline is trained on the exact test distribution rather than handicapped. In the static comparison it receives 2,000 batched training steps; in the stream protocol it receives 200 steps on each newly arrived pattern, with no replay buffer, mirroring the cell's one-pass condition.

3.3 Protocols

Static recall (Engine 23A): 25 patterns stored jointly; recall from fragments with keep $\in \{0.50, 0.25\}$. Stream (Engines 23B, 24): patterns arrive one at a time and are never revisited; after the full stream, every pattern is probed from fragments, with retention reported separately for the first ten and last ten written. The classical cell is run both under capacity ($K = 30 < 0.138N \approx 35$) and deliberately over it ($K = 60$) to expose its failure wall honestly. The sharpened cell is run at $K = 60, 200$ and 1000 — the latter being four memories per unit — and the empirical minimum fragment for $\geq 95\%$ recall is swept at K up to 4,000. Balance protocols: Engine 18 sweeps the residual gain α in $h \leftarrow h + \alpha \cdot f(h)$ for depths $L \in \{8, 16, 32\}$ in trained residual networks under long (1,200-step) and tight (200-step) budgets; Engine 23C sweeps the continuous relaxation gain of the classical cell. All runs are seeded; engine scripts accompany this paper.

4. Results

4.1 The static race is won by the sled — recorded

When the feedforward baseline is granted task-specific gradient training (9.4×10^9 multiply-accumulates), it beats the classical cell at static fragment recall: at keep = 0.50, sled success 89.3% against cell 60.0%; at keep = 0.25 both fail (1.3% and 5.3%). The expansion architecture, given its full training budget on groomed conditions, wins the drag race. The framework does not predict otherwise; the wheel's claim is structural, not universal, and this negative is preserved.

Condition	Cell overlap	Cell success	Sled overlap	Sled success
keep = 0.50	0.825	60.0%	0.969	89.3%
keep = 0.25	0.362	5.3%	0.550	1.3%

Table 1. Engine 23A — static recall, $K = 25$, joint storage/training, matched parameters.

4.2 The stream: writing without erasing versus learning by erasing

The stream protocol is where the structural difference lives, because the gradient-trained network must disturb the weights that hold patterns 1..k in order to learn pattern k+1, while the Hebbian fold deposits without erasing. Under capacity ($K = 30$), the classical cell recalls 66.7% of all patterns from half-fragments after the full one-pass stream and holds its first ten written at 70.0%; the sled collapses to 14.4% overall with its first ten at exactly 0.0% — total catastrophic forgetting — remembering only the recent tail (50.0% on the last ten). The colloquial statement of the framework — you cannot unlearn something — is here a measured property of the writing rule. Over capacity ($K = 60$), however, the classical cell does not degrade; it shatters to 0.0% on everything at once. This is the known blackout catastrophe at 0.138N and it is the classical cell's disqualifying defect as a memory organ.

K	System	Overall	First-10	Last-10	Note
30	Cell (classical)	66.7%	70.0%	43.3%	under capacity
30	Sled	14.4%	0.0%	50.0%	catastrophic forgetting
60	Cell (classical)	0.0%	0.0%	0.0%	blackout wall at 0.138N
60	Sled	6.7%	0.0%	36.7%	recency only

Table 2. Engine 23B — one-pass stream, no replay, recall from keep = 0.50 fragments.

4.3 Sharpening the collapse removes the wall

Engine 24 replaces the quadratic energy with the dense (softmax-separated) form and re-runs the stream far beyond the classical wall. The wall is gone. At $K = 60$, where the classical cell scored zero, the dense cell recalls 100% from half-fragments. At $K = 1,000$ — four memories per unit — it still recalls 100% from half-fragments,

settling in two to three steps, and the first ten memories written recall exactly as well as the last ten: zero forgetting across the entire stream, with every write a one-row append that disturbs nothing. The matched sled, run at $K = 60$ under the same one-pass protocol, scores 3.9% overall with first-10 at 0.0%.

K	keep	Overlap	Success	Steps	First-10	Last-10
60	0.50	1.000	100.0%	2.1	100.0%	100.0%
60	0.25	0.973	97.2%	3.0	90.0%	93.3%
200	0.50	1.000	100.0%	2.3	100.0%	100.0%
200	0.25	0.888	88.0%	3.1	96.7%	86.7%
1000	0.50	1.000	100.0%	2.6	100.0%	100.0%
1000	0.25	0.828	81.7%	3.3	70.0%	56.7%

Table 3. Engine 24 — dense cell, one-shot writes, recall over the full stream. Sled at $K = 60$ (same protocol): 3.9% overall, first-10 = 0.0%.

The honest scaling boundary behaves as the separation argument predicts: the cue must beat the maximum of K noise overlaps, which grows only as $\sqrt{2 \ln K}$, so the minimum usable fragment creeps upward slowly with library size — 0.30 at $K = 60, 200$ and 1,000, rising to 0.40 at $K = 4,000$. A larger library does not forget; it demands slightly better questions.

Library size K	Minimum fragment for $\geq 95\%$ recall	Memories per unit
60	0.30	0.23
200	0.30	0.78
1000	0.30	3.9
4000	0.40	15.6

Table 4. Engine 24 — empirical fragment-quality boundary versus library size ($N = 256$).

4.4 The energy ledger

At matched parameter count and identical task, learning 25 memories cost the classical cell 1,638,400 multiply-accumulates (one local fold each) against 9,437,184,000 for the gradient-trained sled: a measured ratio of 5,760x. In the stream protocol the per-memory comparison is starker still: the dense cell's write is an append of one pattern (≈ 256 operations, zero disturbance) against approximately 3.1×10^8 multiply-accumulates per memory for sequential gradient training — roughly six orders of magnitude — and the gradient learner erases its own past in the process. This is the framework's balance-not-throttle claim rendered as arithmetic, and it is consistent with Landauer's placement of the thermodynamic cost on erasure: the writing rule that never erases is also the one that costs almost nothing.

Operation	Cell	Sled
Learn 25 memories (classical comparison)	1.64×10^6 MACs	9.44×10^9 MACs
Write one memory (stream, dense form)	≈ 256 ops, no disturbance	$\approx 3.1 \times 10^8$ MACs, erases prior
Recall one query (dense, $K = 1000$)	$\approx K \cdot N$ per step, 2–3 steps	one forward pass

Table 5. The energy ledger. Learning is where the $\times$ orders of magnitude live; recall costs are comparable.

4.5 Where H does not live — two preserved negatives

The framework hypothesizes $H = \pi/9 \approx 0.349$ as a universal feedback optimum. Two dynamical protocols tested this directly and both returned negatives, which are recorded with the confirmations. Engine 18 swept the residual gain α in trained deep residual networks ($h \leftarrow h + \alpha \cdot f(h)$) at depths 8, 16 and 32 under long and tight step budgets. Under the long budget the loss is monotone in α with no interior optimum. Under the tight budget — where adaptation speed matters — an interior optimum does emerge, and the framework’s qualitative two-walled prediction (stagnation below, instability above) is confirmed; but the optimum is not pinned at H. It scales with depth as $\alpha^* \approx 2.5/\sqrt{L}$ (measured $\alpha^* \cdot \sqrt{L} = 2.40, 2.60, 2.55$ across a fourfold depth range), crossing 0.349 only near $L \approx 52$ — a depth intriguingly near SHA-256’s 64 rounds, noted here as an open curiosity and nothing more. Engine 23C swept the relaxation gain of the classical cell’s continuous collapse dynamics and found a clean Pareto frontier — recall quality rising monotonically as the gain falls, convergence speed rising as it climbs — with no distinguished interior point; H sits on the frontier (overlap 0.952 at 71 steps) but is not selected by it. Under the protocols tested, balance is a tradeoff curve, not a universal constant. If H has a seat in this machinery, these results constrain it to step-budgeted or depth-specific formulations, which remain open.

5. Interpretation Under the Framework

Read through the NEXUS ontology, the results assemble as follows. Writing is a fold: the Hebbian deposit and the dense append both add constraint to a field without destroying what is there — bits are flipped into structure, not erased, and the measured zero-forgetting at a thousand one-shot writes is that conservation made visible. Recall is the collapse: clamp the fragment (fix C) and let the free units adjust (a and b) until the configuration closes on the attractor the cue already implied. Nothing is searched; the state falls, in two to three settling steps, into the only basin consistent with the constraints — the protein’s move, not Levinthal’s. The energy asymmetry follows from the direction of the arrow: the expansion architecture pays its enormous bill at learning time because gradient descent is throttle — global, iterative, erasive — while the collapse architecture pays almost nothing because deposition is local and final.

The sharpest interpretive fact is the published identity between the dense cell’s settling step and transformer attention. The one component of modern AI that practitioners describe as doing the inexplicable work — attention — is, verbatim, a single collapse of this memory. On this reading the transformer is a hybrid: a collapse organ (attention, reading structure out of context in one settle) embedded in an expansion chassis (the feedforward stack, the gradient learner) that re-expands after every collapse and pays the energy bill.

The framework's critique is thereby located precisely: the waste is not in the part that thinks; it is in the chassis around it. This also connects to the author's v55 Triadic Cell result, in which deterministically compiled structure (the NeedSlot contract and gated evidence fold) lifted a 1.5-billion-parameter model from 0.90625 to a perfect 1.0 on a 96-sample adversarial benchmark with zero induced errors: there as here, structure substitutes for scale, and the intelligence improves when the model is made to operate inside compiled constraint rather than to free-run.

6. Limitations, Stated Plainly

The experiments are toy-scale by design: $N = 256$ units, random bipolar patterns, not language or perception. Random patterns are the favorable case for associative memory; structured, correlated patterns (real session embeddings) will lower capacity and constitute the necessary next test. The sled comparisons establish the structural contrast, not superiority over the continual-learning literature: methods such as rehearsal buffers and elastic weight consolidation mitigate forgetting in gradient systems and were not benchmarked here. The dense cell's storage grows linearly with memories and its recall cost grows as $K \cdot N$ per settling step; under the framework the growth is the thesis, but at scale it implies hierarchical organization (collapse over collapses) that is proposed, not demonstrated. The attention identity is established literature used interpretively. And the H-hypothesis results are negative under the protocols tested; they do not, and cannot, rule out a seat for the constant under protocols not yet run. None of these caveats is decorative; each marks the next engine.

7. Open Problems and Next Engines

Four bolts are exposed and ordered. First, structured latents: replace random patterns with embeddings of the author's actual research corpus (12,190 session windows are already prepared) and measure fragment-recall on real, correlated structure; the NeedSlot schema of the v55 architecture is the candidate for giving the latent a forced internal geometry. Second, hierarchy: cells whose memories are themselves the settled states of lower cells — collapse over collapses — as the structural answer to linear storage growth. Third, the seat of H: step-budgeted and criticality-style protocols for the settling dynamics, since the tight-budget protocol of Engine 18 was the one that produced an interior optimum at all. Fourth, hardware: the settling operation is what analog and memristive substrates do natively in physics-time; the six-orders-of-magnitude learning ledger measured here in software is the prediction that, in such a substrate, the energy claim becomes physical rather than arithmetical. Each is falsifiable; each will be run under the same discipline as the engines reported here.

8. Conclusion

Built as a collapse engine rather than an expansion engine, a parameter-matched memory wrote a thousand one-shot memories without forgetting its first, learned at a cost five thousand seven hundred and sixty times smaller than its gradient-trained rival in the directly matched comparison, and recalled by settling from fragments in two steps — while the rival, structurally obliged to erase in order to learn, lost its earliest memories completely. The classical form of the cell carries a catastrophic capacity wall, removed by sharpening the collapse; the sharpened form's settling step is identically the attention operation, locating the collapse primitive inside the architecture it critiques. Two protocols designed to find $H = \pi/9$ as a universal feedback constant instead found a depth-scaled law and a Pareto frontier, and those negatives are preserved

as constraints on the theory. If we figure out part of this, we get all of it — but only by this route: one forced result at a time, negatives kept, structure first.

Acknowledgments and Reproduction

Engines 18, 20, 23 and 24 were designed in dialogue and executed live by Claude (Anthropic) under the author's NEXUS constraints and session standards; all numbers in this paper are taken verbatim from those seeded runs, and the engine scripts (`engine18_gain_sweep.py`, `engine18b_tight_budget.py`, `engine20_slot_tests.py`, `engine23_nexus_cell.py`, `engine24_dense_cell.py`) accompany this document for independent reproduction. The v55 Triadic Cell results cited in Section 5 are reported in the author's companion paper, *Operational Geometry and Compiler-Rooted Slot Control* (May 2026).

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